

Herald



Herald — Issues

Field	Value
Revision	15
Created	2026-05-20
Last modified	2026-05-22
Status	active
Status summary	r15 captures Wave 6 (pherald inbound runtime + Claude Code headless bridge — code-doc closure; tag v0.4.0 deferred to T13b post live evidence). HRD-100 atomic Issues→Fixed in this revision. New Cobra subcommand <code>pherald listen</code> runs the closed-loop runtime: Telegram <code>getUpdates</code> long-poll (<code>OnText/OnPhoto/OnDocument/OnVoice</code>) + bot self-filter (<code>bot.Me.Username</code> anti-echo guard, fails <code>Subscribe</code> if <code>getMe</code> returned no <code>username</code>) + sha256 content-addressed attachment download into <code>~/ .herald/ inbox/<sha256>.<ext></code> + Claude Code dispatch with Opus pinned in argv (<code>claude --model claude-opus-4-7 --resume <UUID> --print <envelope></code>) + envelope pre-text (verbatim operator wording "We have received new message from our communication channel") preceding the <code><<<HERALD-DISPATCH-v1>>></code> structured block + <code><<<HERALD-REPLY>>></code> parser with action routing (<code>reply / issue.open / event.emit</code>) + <code>tgram.Adapter.SendReply</code> wiring <code>reply_to_message_id</code> from the inbound <code>message_id</code>. T10a --qa-out-dir flag journals every inbound/outbound event as JSONL per §107.x. 12 commits (T1=ad87d7f..T12=96c7c6b). 8 new e2e invariants E63-E70 added (currently SKIP with documented reasons — converting to PASS at T10b when operator runs the live closed-loop and commits <code>docs/qa/HRD-100/</code>). Wave 6 mutation gate <code>tests/test_wave6_mutation_meta.sh</code> lands 3 paired hermetic mutations. Tag v0.4.0 deferred to T13b (operator-supplied live evidence per §107.x). Prior r14 captured Wave 4b TOON <code>application/toon</code> content negotiation (tag v0.3.0). Prior r13 captured Wave 4a (HTTP/3 + Brotli + Alt-Svc + TLS 1.3). r12 closed HRD-016 (Wave 3b pherald Runner live). r11 closed HRD-011 (Telegram live, <code>message_id=5</code>).
Issues	HRD-008, HRD-015, HRD-018..HRD-027, HRD-029..HRD-056, HRD-081, HRD-085..HRD-090
Issues summary	47 open / in-progress workable items. r15: HRD-100 closed atomically (Wave 6 pherald inbound runtime — code-doc closure; live evidence operator-pending → T10b). r14: substrate upgrade only (Wave 4b TOON; no HRDs migrated). r12:

Field	Value
	HRD-016 closed atomically (Wave 3b pherald Runner live + e2e E37-E42 PASS or honest SKIP-with-reason when PG :24100 absent). r11: HRD-011 (Telegram live) closed atomically with live message_id=5 evidence. r10 doc-cleanup renumbered HRD-093 → HRD-099. r9: HRD-028 + HRD-098 closed (Wave 3a). Carry-over to Wave 3c: HRD-024 (iherald paging), HRD-033 (destructive-guard body), HRD-018..027 flavor-specific bindings, plus the rest of the §43 command catalogue.
Fixed	HRD-001..HRD-007, HRD-009, HRD-009b, HRD-010, HRD-011, HRD-012, HRD-013, HRD-014, HRD-016, HRD-017, HRD-028, HRD-080, HRD-092, HRD-093, HRD-094, HRD-095, HRD-096, HRD-097, HRD-098, HRD-099, HRD-100, HRD-110, HRD-111, HRD-112, HRD-113, HRD-114
Fixed summary	see Fixed.md.
Continuation	see CONTINUATION.md.

Table of contents

- [Open](#)
- [In progress](#)
- [Blocked](#)
- [Conventions](#)

Open

Per Universal §11.4.74 every new row carries a Catalogue-Check line in its References cell (one of reuse|extend|no-match <org/repo>@<sha>). For HRDs whose catalogue-check is still pending, the row records no-match (2026-05-20) — operators reviewing the implementation PR will re-verify.

ID	Type	Status	Criticality	Title	Opened
HRD-015	task	open	low	Inheritance gate I8 invariants for Go scaffold (go.work + commons/types.go + null adapter test passes)	2026-05-20
HRD-018	task	in_progress	high	commons_constitution Go package: Evaluator interface + 12 event-class emit helpers + constitution_state + constitution_bindings table migrations + bundle-hash captureer + mode-ladder runtime config	2026-05-20
HRD-019	task	open	high	cherald constitution bindings — bulk implementation: ~30 .policy.violation rules + .gate.failed/.gate.recovered/.credential.leak/.sp handlers; /v1/compliance pull surface	2026-05-20
HRD-020	task	open	high	sherald host-safety + repo-safety bindings (§9.1/.2/.3, §12.1/.2/.3/.6, §11.4.32, §11.4.36, §11.4.41, §11.4.71) — destructive-op detection hook, force-push interceptor, mem-budget watcher	2026-05-20

ID	Type	Status	Criticality	Title	Opened
HRD-021	task	open	middle	bherald CI/test bindings (§1, §11.4.2/.3/.4/.5/.7/.13/.14/.24/.27/.39/.43/.46/.48–.52/.67) — gate-result event emitters; integrates with the consuming project's CI workflow	2026-05-20
HRD-022	task	open	middle	rherald release bindings (§4 tag mirroring, §5 changelog, §11.4.38 installable-asset evidence, §11.4.40 full-suite retest)	2026-05-20
HRD-023	task	open	middle	pherald project bindings (§2 commit+push, §3 submodule propagation, §11.4.11/.15/.21/.22/.34/.37/.42/.55/.66/.71/.74)	2026-05-20
HRD-024	task	open	middle	iherald constitution-rule escalation bindings (§11.4.10/.10.A credential-leak page-out, §11.4.21 + §11.4.66 operator-blocked escalation)	2026-05-20
HRD-025	task	open	low	scherald scheduled-audit bindings (§11.4.45 periodic Status.md sweep + daily/weekly/monthly compliance digest)	2026-05-20
HRD-026	task	open	middle	Constitution-bundle hash captureer — computes SHA-256 of rendered Constitution.md at evaluation time; persists on every emitted event for replayability	2026-05-20
HRD-027	task	open	middle	Mode-ladder runtime config (constitution_bindings table + admin REST endpoints to flip allow/warn/enforce per binding per tenant without redeploy)	2026-05-20
HRD-029	task	open	middle	§2 pherald commit-push — single-entripoint locked commit + multi-mirror push	2026-05-20
HRD-030	task	open	middle	§3 pherald submodule propagate — owned-submodule walk in propagation order	2026-05-20
HRD-031	task	open	middle	§4 rherald tag mirror — assert tag exists on every owned submodule	2026-05-20
HRD-032	task	open	low	§5 rherald changelog generate — Conventional Commits → docs/changelogs/<v>.md + multi-format export	2026-05-20
HRD-033	task	open	high	§9.1 sherald destructive guard <op> — wrap rm/git-reset/git-push-force with prerequisite checks	2026-05-20
HRD-034	task	open	middle	§9.3 sherald backup snapshot <path> — hardlinked snapshot helper	2026-05-20
HRD-035	task	open	middle	§11.4.2 bherald evidence capture <test_id> — captured-evidence recorder	2026-05-20
HRD-036	task	open	high	§11.4.10 cherald creds scan — gitleaks/trufflehog integration, emits .credential.leak	2026-05-20
HRD-037	task	open	low	§11.4.12 cherald docs sync — regen Issues_Summary / Fixed_Summary / Status_Summary	2026-05-20
HRD-038	task	open	low	§11.4.18 cherald script-docs check — assert sibling .md for every scripts/**/*.*.sh	2026-05-20
HRD-039	task	open	low	§11.4.19 / .23 cherald fixed align + cherald colorize — Issues/Fixed format + HTML colorizer	2026-05-20
HRD-040	task	open	high		2026-05-20

ID	Type	Status	Criticality	Title	Opened
				§11.4.26 sherald constitution pull — wrap fetch + rebase + validation gate, emits <code>.bundle.updated</code>	
HRD-041	task	open	middle	§11.4.27 bherald test-tier verify — 8-tier matrix verification	2026-05-20
HRD-042	task	open	low	§11.4.31 cherald submanifest verify — Submodule-Dependency-Manifest gate	2026-05-20
HRD-043	task	open	high	§11.4.36 pherald install-upstreams — install_upstreams wrapper	2026-05-20
HRD-044	task	open	middle	§11.4.37 pherald fetch-guard — pre-edit fetch + rebase enforcement	2026-05-20
HRD-045	task	open	high	§11.4.40 rherald gate retest — pre-tag full-suite retest gate	2026-05-20
HRD-046	task	open	high	§11.4.41 sherald force-push gate — merge-first + per-session-auth enforcement	2026-05-20
HRD-047	task	open	middle	§11.4.45 / .56 scherald status digest — periodic Status.md sweep + Status_Summary regen	2026-05-20
HRD-048	task	open	low	§11.4.53 cherald fixed-summary sync — standalone Fixed_Summary backfill	2026-05-20
HRD-049	task	open	middle	§11.4.55 pherald reopen <HRD-NNN> — Issues→Fixed reversal + Reopens history	2026-05-20
HRD-050	task	open	middle	§11.4.59 cherald readme sync — README doc-link regen + multi-format re-export	2026-05-20
HRD-051	task	open	high	§11.4.60 cherald composite-gate — canonical implementation of CM-DOCS-COMPOSITE-SYNC	2026-05-20
HRD-052	task	open	middle	§11.4.65 cherald export — bulk Markdown export wrapper (md/html/pdf/docx)	2026-05-20
HRD-053	task	open	high	§11.4.71 pherald pre-push — fetch + investigate + integrate hook	2026-05-20
HRD-054	task	open	middle	§11.4.73 cherald spec-version check — Revision-vs-edits drift detection	2026-05-20
HRD-055	task	open	middle	§11.4.74 cherald catalogue-check <pr> — scan PR for Catalogue-Check + survey runner over vasic-digital + HelixDevelopment	2026-05-20
HRD-056	task	open	high	§12.6 sherald mem-budget watch — daemon-mode 60% threshold watcher emitting <code>.host.safety_breach</code>	2026-05-20
HRD-081	task	open	low	Extend <code>digital.vasic.containers/pkg/compose</code> to detect podman-compose vs docker compose runtime and either emit runtime-appropriate flags (no <code>--wait</code> for podman-compose) or fall back to host-side healthcheck polling. Composes with <code>§11.4.76</code> — required so <code>compose.WithWait(true)</code> works across both backends. Also fix <code>Status()</code> parsing for podman-compose ps output (returns 0 services even when containers are visible to podman directly). Per §11.4.76: extend upstream submodule, never reimplement.	2026-05-20

ID	Type	Status	Criticality	Title	Opened
HRD-085	task	open	middle	Implement single-task admin operations on <code>commons_infra.pgxTaskRepository</code> — <code>GetByID</code> , <code>Update</code> , <code>Delete</code> . Required for <code>Requeue + MoveToDeadLetter</code> call chains and operator tooling. Currently all three return <code>ErrUnsupported</code> (HRD-085). Scope: read one <code>BackgroundTask</code> row by ID; full-row <code>Update</code> including JSONB columns; soft- or hard- <code>Delete</code> decision per §107.	2026-05-20
HRD-086	task	open	middle	Implement worker-side running-task state mutations on <code>commons_infra.pgxTaskRepository</code> — <code>UpdateStatus</code> , <code>UpdateProgress</code> , <code>UpdateHeartbeat</code> , <code>SaveCheckpoint</code> . Required by River workers + the stuck-detector reaper. Currently all four return <code>ErrUnsupported</code> (HRD-086). Scope: status transitions guarded by valid-state-machine check; progress as <code>(float64, message)</code> ; heartbeat as <code>updated_at = now()</code> writeback; checkpoint as opaque <code>[]byte</code> blob persisted alongside the task row.	2026-05-20
HRD-087	task	open	low	Implement admin / stats / audit reads on <code>commons_infra.pgxTaskRepository</code> — <code>GetByStatus</code> , <code>GetPendingTasks</code> , <code>CountByStatus</code> , <code>GetTaskHistory</code> . Drives the <code>/v1/queue/*</code> introspection endpoints + <code>Peek</code> delegate. Currently all four return <code>ErrUnsupported</code> (HRD-087). Scope: status-filtered paginated reads; pending-task slice for <code>Peek</code> ; status-bucketed counts; per-task execution-history rows from the <code>task_execution_history</code> table.	2026-05-20
HRD-088	task	open	middle	Implement queue-recovery reads on <code>commons_infra.pgxTaskRepository</code> — <code>GetStaleTasks</code> , <code>GetByWorkerID</code> . Drives the stuck-task reaper + worker-crash recovery loop. Currently both return <code>ErrUnsupported</code> (HRD-088). Scope: stale-threshold filter against <code>updated_at</code> ; per-worker assignment lookup for re-dispatch after a worker disappears.	2026-05-20
HRD-089	task	open	low	Implement resource-monitor I/O on <code>commons_infra.pgxTaskRepository</code> — <code>SaveResourceSnapshot</code> , <code>GetResourceSnapshots</code> . Persists per-task CPU/memory/IO samples for capacity-planning + post-mortem. Currently both return <code>ErrUnsupported</code> (HRD-089). Scope: append-only insert keyed by <code>(task_id, captured_at)</code> ; per-task paginated read in reverse-chronological order. Schema target: a <code>task_resource_snapshots</code> table not yet in the §9.6 migration set — adding it is part of this HRD.	2026-05-20
HRD-090	task	open	middle	Implement dead-letter handling on <code>commons_infra.pgxTaskRepository</code> — <code>MoveToDeadLetter</code> . Required failure path when a	2026-05-20

ID	Type	Status	Criticality	Title	Opened
HRD-131	task	open	middle	task exhausts retries or fails a §107 invariant. Currently returns <code>ErrUnsupported</code> (HRD-090). Scope: atomically move the <code>BackgroundTask</code> row to a <code>dead_letter_tasks</code> table with the failure-reason string + original-row JSONB snapshot; emit a <code>.queue.dead_letter</code> event via <code>commons_constitution</code> per §42.1. Schema target: a <code>dead_letter_tasks</code> table not yet in the §9.6 migration set — adding it is part of this HRD. Migrate text HRD trackers to versioned SQLite single-source-of-truth (§11.4.93 / Herald §108.h, 6-phase) — DB committed as authoritative SSoT per operator mandate 2026-05-27. Phase 1 (this item): file the scope-locked tracking HRD. Phase 2: scaffold/adopt the migration Go binary that already lives in the constitution submodule (<code>constitution/scripts/workable-items/</code>) — Herald references it per §11.4.74 catalogue-first and never reimplements it. Phase 3: sync <code>md→db</code> (load <code>docs/Issues.md / docs/Fixed.md / *_Summary.md / docs/Status.md / docs/CONTINUATION.md</code> §3 into <code>docs/.workable_items.db</code>). Phase 4: sync <code>db→md</code> byte-identical (round-trip regeneration so MD↔DB drift is mechanically impossible). Phase 5: generator shims (the <code>*_Summary.md</code> variants + <code>Status/CONTINUATION</code> become derived from the DB). Phase 6: prohibit text-direct edits (release-gate guard that the DB is the only authoring surface). Operator divergence from the parent §11.4.93 gitignored-with-regeneration default: for Herald the DB IS the authoritative artefact — <code>docs/.workable_items.db</code> is version-controlled (committed), NOT regenerated-on-clone; only the transient WAL/SHM sidecars (<code>docs/.workable_items.db-wal / -shm</code>) stay gitignored ("We should not git ignore our workable items database since it is our single source of truth regarding items we have to do on the project!"). Scope-lock: this HRD covers Phase 1 (this row) + Phase 2 adoption ONLY. Phases 3-6 (<code>md→db</code> sync, <code>db→md</code> byte-identical regeneration, generator shims, text-direct-edit prohibition) are follow-on work-items to be opened as Phase 2 completes, so the §106 spec-change / §107.x evidence invariants are never broken mid-migration. §11.4.85 stress+chaos shared scaffold — Go-native <code>commons/stresschaos</code> test-helper package (GAP-3 plan §2 / §6 T1). Provides: a bounded concurrent load-runner (<code>RunLoad</code> — N workers × M iterations of a closure via <code>sync.WaitGroup</code>, no new dependency: golang.org/x/sync/errgroup is NOT in <code>go.mod</code> so <code>stdlib sync</code> is used per the plan's "prefer NOT adding deps" tooling note); nearest-rank percentile	2026-05-27
HRD-122	task	in_progress	middle		2026-05-27

ID	Type	Status	Criticality	Title	Opened
HRD-123	task	in_progress	high	math (Percentiles → {p50,p95,p99,max,min,count}) emitted as latency.json+latency_histogram.csv; an evidence-dir writer creating docs/qa/<run-id>/stress_chaos/<surface>; and a best-effort host-memory headroom probe (HostMemHeadroom — vm_stat+sysctl hw.memsize on darwin, /proc/meminfo on linux) for the §12.6 60%-ceiling proof (degrades to probe-unavailable, never a hard failure). Self-tested: 8 unit tests (percentile correctness on a known 1..100ms dataset, evidence-dir creation, bounded-goroutine proof, host-mem best-effort, §12.6 ceiling predicate) PASS under go test -race -count=1. Lowest sensible layer per CLAUDE.md layering (commons L0, importable by every flavor). §11.4.85 Gin /v1/events stress+chaos tests (GAP-3 plan §1 row 1 / §6 T2) — new pherald/internal/http/events_stress_chaos_test.go driving the REAL POST /v1/events handler (EventsHandler → runner.Run, all 7 stages) over a REAL in-process http test server dialed by net/http.Client, through the production Gin middleware chain (cli.TOONMiddleware + auth gate). Only the external boundary (PG/Redis/channel via runner.NewFakeRunner) is faked per §11.4.27; the handler, body read+transcode, error→status mapping, and orchestration run unmodified. STRESS: N=12 workers × M=200 = 2400 CloudEvent POSTs, -race clean — 0 errors, 0 5xx, all 202; p50=0.706ms/p95=1.755ms/p99=2.665ms/max=5.722ms, ~13,322 req/s. CHAOS: (b) duplicate-key-under-load (1200 POSTs, per-worker keys) → 12×202 + 1188×200, 0 5xx, 0 hang — graceful degrade; (c) input-corruption (7 bodies: 8 MiB random, truncated JSON, non-UTF8, JSON array, missing-id, garbage, empty) × 5 reps → every body a tagged event_parser: 400, 0 panic, 0 5xx; (d) auth storm — 1000 random 32-byte bearer tokens through the REAL commons_auth.GinMiddleware+HMAC(HS256) Verifier → 1000×401, 0 bypass (no token minted → no new go.mod dep); (a) live PG-drop (container pause) → SKIP-with-reason (no runtime; fail-loud-on-PG-error proven hermetically at runner layer HRD-125). Captured evidence under docs/qa/HRD-123-stress-chaos-2026-05-27T132755/stress_chaos/events/ (latency.json, latency_histogram.csv, throughput.csv, redis_down_fallback.log, categorised_errors.txt, auth_storm.log, recovery_trace.log, host_memory_headroom.txt, summary.md). FINDING (→ HRD-132 confirmed at HTTP layer): a SHARED-	2026-05-27

ID	Type Status	Criticality Title	Opened
HRD-125 task	in_progress high	key cross-worker concurrent replay flood triggers the REAL runner . go : 132 CachedRcpt . WasReplay data race — the NewFakeRunner events_processed store caches+shares the *Receipt, so concurrent replays mutate it (this IS the "activates with Wave 4+ Receipt caching" case HRD-125 predicted). Test uses per-worker keys (production-faithful: real PG does NOT cache the Receipt today) so the HTTP-surface degrade contract is proven - race-clean without asserting a property the code cannot honour (= §107 PASS-bluff); the race is OWNED by HRD-132. Host-mem headroom recorded; HRD-123 is in-process (not a resource-exhaustion surface) so the §12.6 ceiling gate does not gate it. §11.4.85 Runner 7-stage pipeline stress+chaos tests (GAP-3 plan §1 row 3 / §6 T4) — new pherald / internal / runner / stress_chaos_test . go exercising the REAL runner . Run (only the external boundary — PG events_processed store, channel sink, Redis SETNX seam — is faked per §11.4.27; the orchestrator + 7 stages run unmodified). STRESS: N=16 goroutines × 40 iters concurrent same-key replay + fresh keys, - race clean (PASS). CHAOS: (a) 1000× duplicate flood @ 50 parallel under live atomic Redis SETNX; (a') same flood under the real nil-Redis PG-fallback degrade; (b) hermetic PG-deadlock (SQLSTATE 40P01) injection on the events_processed Insert. Captured evidence under docs / qa / HRD - 125 - stress - chaos - 2026 - 05 - 27T131251 - gap3 / stress_chaos / runner / (latency.json p50=0.252ms/p95=1.075ms/p99=2.966ms, histogram CSV, race_clean.log, exactly_once.txt, duplicate_flood.txt, nil_redis_degrade.txt, deadlock_recovery.log, host_memory_headroom.txt, summary.md). FINDING (→ HRD-132): dispatch exactly-once is NOT guaranteed under concurrent replay — the events_processed archive row is written at Stage 7 (end of pipeline), so the Stage-2-fresh-verdict → Stage-7-archive-commit window admits a bounded handful of concurrent duplicate dispatches (observed 2–4 of a 1000× flood; never the full 1000). Archival (events_processed) IS exactly-once (PG UNIQUE + ON CONFLICT DO NOTHING). Tests assert the honest code-true contract (archival exactly-once + bounded dispatch), NOT an aspirational sends==1 (that would be a §107 PASS-bluff). A latent runner . go : 132 CachedRcpt . WasReplay data race surfaces only when a store caches+shares a *Receipt (current production PG Lookup returns Receipt=nil → no race today; would activate with the planned Wave 4+ Receipt caching).	2026-05-27

ID	Type	Status	Criticality	Title	Opened
HRD-132	bug	open	high	Tighten Runner idempotency window so concurrent same-key replay cannot duplicate-dispatch a notification (FINDING from HRD-125 stress/chaos). <code>idempotency.go:64-72</code> treats a Redis-SETNX-miss-with-no-PG-row as FRESH (the documented §32.1 "Redis-lies-PG-truths" design, to avoid ingest lockup), but the <code>events_processed</code> archive row is only written at Stage 7 (<code>OutcomeRecorder.Process</code>) — so concurrent same-key requests arriving in the Stage-2→Stage-7 window ALL dispatch (observed 2–4 of a 1000× flood; never the full 1000). Archival is exactly-once (PG UNIQUE + ON CONFLICT DO NOTHING); channel <i>dispatch</i> is only bounded → for a notification system this is duplicate-alert delivery under concurrent load. Fix direction: claim the idempotency slot atomically BEFORE dispatch (<code>INSERT events_processed ... ON CONFLICT DO NOTHING</code> at Stage 2, treat 0-rows-affected as duplicate, so the PG row — not Stage-7 — is the dispatch gate), preserving the nil-Redis PG-only degrade (HRD-179). ALSO fix the latent runner <code>.go:132</code> <code>CachedRcpt.WasReplay</code> data race (mutates a possibly-shared <code>*Receipt</code>; dormant today since PG Lookup returns <code>Receipt=nil</code>, activates with Wave 4+ Receipt caching). MUST ship with the §11.4.85 stress test upgraded to assert <code>sends==1</code> under the same 1000× flood (the stronger assertion HRD-125 deliberately could NOT make without this fix — §11.4.4 test-interrupt-on-discovery).	2026-05-27

In progress

ID	Type	Status	Criticality	Title	Opened	Last update	References
HRD-008	task	in_progress	middle	Operator-side quickstart compose validation (Postgres + Redis + OTel + pherald container)	2026-05-20	2026-05-20	spec V3 §26.5 — scaffold shipped, live end-to-end run pending operator; Catalogue-Check: no-match (2026-05-20) — bespoke to Herald

Blocked

(none)

Conventions

See [Fixed.md](#) for closed items + Universal §11.4.12/.15/.16/.19/.33/.55/.73/.74 composition rules.